Model training investigation

Authorized three-hour window | 28 October 2026 | Branch mohammad

Stronger construction, with a limited fresh-map learning gain.

The old primitive policy averaged 0.6 trips on ten fresh maps. The planner-aided frozen initial policy averaged 119.1; the selected trained goal policy averaged 125.7. The secondary sampled-policy means were 130.4 and 130.4. Most of the change from the old agent comes from the new route-planning/action abstraction and informed starting policy. The initial-versus-trained comparison is the relevant check for learning; one small experiment cannot establish robust convergence.

Measurement	Outcome
Best recorded primitive training game	126 trips
Best recorded planner-aided training game	356 trips
Planner-aided frozen start, development mean	162.6 trips (5 maps)
Selected planner-aided checkpoint, development mean	162.6 trips (5 maps)
Target	1,000 actual completed trips; not reached
Rules	OriginalReference; normal finite materials; future schedules hidden

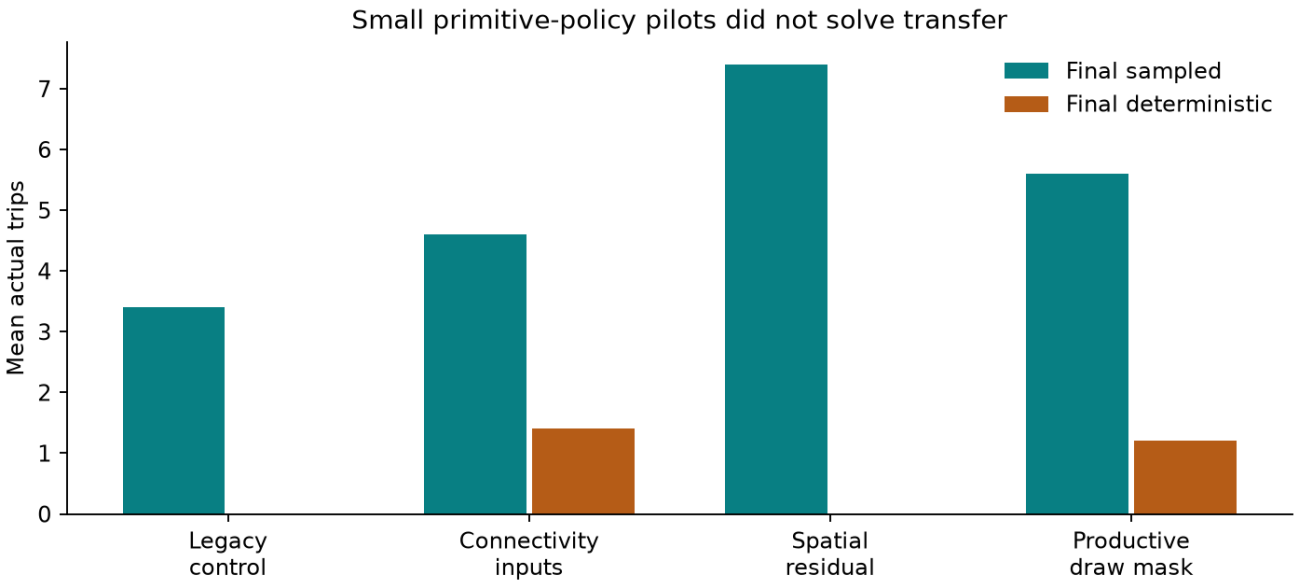
What counts as learning?

A better hand-written prior or a pathfinder can improve scores before any training. Learning means the updated policy improves on its own frozen starting policy on separate maps. This report shows those baselines explicitly. A single impressive replay or increasing shaped reward is insufficient.

The original all-tools agent remains available. The new goal agent is a separate restricted, planner-aided experiment. OriginalReference is a reconstruction, not a validated exact copy of the commercial game. Checkpoints and binaries stay local; small evidence and report sources are archived in Git.

1 / What was wrong?

The main difficulty was coordinating useful road construction, not an artificial game-per-second limit.



Pilot	Selections	Best training	Final deterministic mean
Legacy control	65,536	34	0.0
Connectivity inputs	65,536	33	1.4
Spatial residual	63,490	39	0.0
Productive draw mask	65,536	65	1.2

The input-only pilot preserved starting action probabilities and compared against the same old weights with a fresh optimizer. Adding explicit missing connections did not establish an improvement. Early logs showed about 89% of draws made no measured change to road-tile/connection arrays, motivating endpoint diagnostics. Driveway-only changes or pending-removal cancellation are outside that metric; it is not a complete no-effect test.

Five development seeds, one run per configuration. Spatial residual stopped at its 18-minute cap (63,490 selections); others reached 65,536. Sampled and deterministic policies differ. These are engineering pilots, not replicated significance tests or speed benchmarks. Shared CPU load and validation overhead affect timings.

2 / Better inputs and masks

These changes help the policy understand the current map without changing the game rules.

Change	What it does / why
52-channel option	Adds unconnected houses and destinations without connected matching-color supply to the original 50 map channels. Binary supply is not a capacity guarantee.
Explicit checkpoint adaptation	Copies old weights and zero-fills the new input weights; tests verify initial action probabilities match. A fresh optimizer is explicit.
Spatial residual option	Shared per-cell scoring supplements global coordinate logits; zero initialization preserves the original starting policy.
Productive draw mask	Removes endpoints only when a read-only simulator query proves a stroke cannot change the road network. Possible changes may still be rejected by normal rules.
Immediate draw diagnostics	Counts unchanged road-tile/connection arrays and distinct endpoint pairs before advancing time. Other state changes are outside this measurement.
Development checkpoints	Periodically freezes inference on seeds 2000-2004; logs actual trips and saves validation-best separately from the best training replay.

Reward remains tied to actual delivery

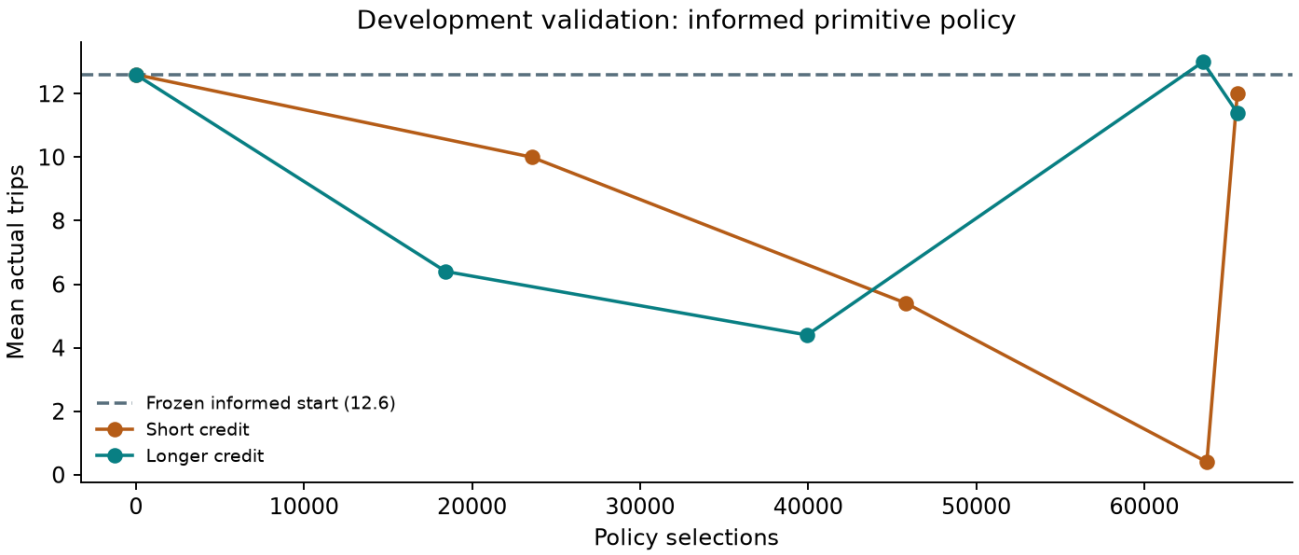
The primitive agent uses completed-trip delta plus discounted potential shaping: reward = new trips + gamma * next potential - previous potential. Potential is 0.25 times the number of usefully connected houses, capped at eight. True game failure resets potential to zero; external time limits retain it for bootstrapping. Breaking and rebuilding a connection cannot repeatedly collect net discounted credit.

All primitive tools remain exposed. Structured inputs exclude future schedules and RNG state. Cells beyond the current playable map are hidden. Every experimental input/action schema is named explicitly so incompatible checkpoints are not silently reused.

Original parameters: 450,610. Connectivity inputs: 450,898. Informed spatial policy: 453,484. Parameter counts describe numerical weights, not games or rules. The original policy has no recurrent memory.

3 / Informed actions and credit

A useful starting bias improved behavior, but that improvement must not be attributed to PPO.



The informed spatial variant initially favors current house-driveway endpoints, matching destination endpoints, waiting when no relevant missing supply remains, and offered upgrades. This is an engineered prior. It does not compute routes. The prior is trainable and the spatial residual can adjust cell scores.

Its frozen deterministic development scores were 0, 0, 8, 33, 22: mean 12.6. The short-credit run used aggregate rollout 256 and GAE lambda 0.95. The longer-credit arm started from the exact same checkpoint, with rollout 1,024 and lambda 0.99. Both used gamma 0.999, four workers, the same reward and productive masks.

Arm	Final deterministic mean	Best training
Short credit	12.0	110
Longer credit	11.4	126

Two interacting credit changes were tested together, not an isolated one-factor ablation. The runs temporarily shared eight simulator processes, so wall speeds are not comparable hardware benchmarks. Selection includes frozen initial and final checkpoints, not only periodic validation-best.

4 / A separate connection agent

The neural policy chooses which connection to build; a visible-state router handles the individual strokes.

Stage	What happens
1. Observe	Read the current map, materials, houses, destination entrances, colors, demand, congestion and connectivity. Never read future spawns.
2. Propose	A deterministic shortest-cost router proposes current matching house/destination connections using ordinary road tiles and available stock.
3. Choose	Maskable PPO scores up to 512 candidate connections, two upgrade choices and waiting. Shared weights score candidate rows rather than memorizing coordinate IDs.
4. Execute	Compress the chosen route into ordinary straight strokes. Each completed command advances normal simulation time. Waiting executes up to five ordinary one-second commands.
5. Learn	Collect actual trip changes plus bounded potential shaping, estimate future returns, and update policy/critic weights with PPO.
6. Check	Freeze the policy on separate development maps and compare against its own frozen initial checkpoint.

What it sees

Each candidate has 25 values: home/destination coordinates and displacement, color, new-road and route costs, demand, missing home/shop supply, existing-road fraction, traffic/waiting, overcrowding, new water tiles and validity. Forty-two globals summarize materials, offers, current clock/week, trips, supply and congestion. Up to 512 candidates are supported; overflow raises an error instead of silently dropping entries.

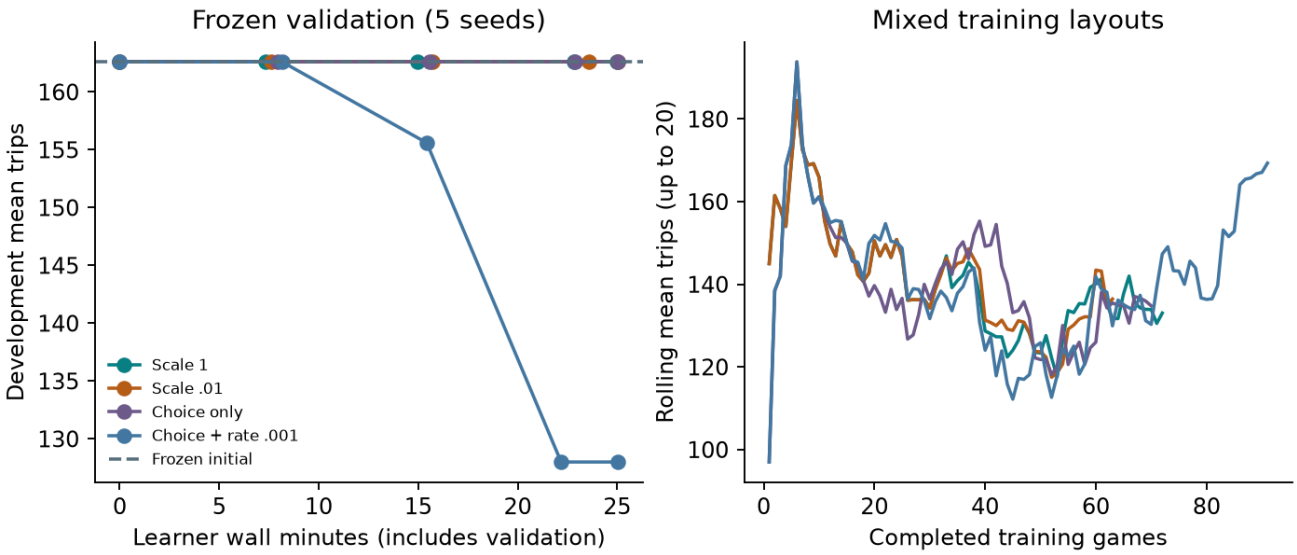
What it cannot do in this experiment

It builds road/bridge strokes, waits and chooses upgrades. Motorways, roundabouts, traffic lights, demolition and manual driveway edits are excluded. This is deliberately a different action abstraction; it does not establish that the original full-tools model learned those skills.

10,334 trainable parameters. Hand-designed initial candidate scores prefer low cost, demand and missing supply. PPO learning rate 0.0001 (final rate pilot: 0.001), entropy 0.005, gamma/lambda 0.99 per macro decision, rollout 256, four epochs. Variable-duration macro actions change the discounting problem. The router is assistance, not learned route construction or an optimal planner.

5 / Did the connection agent learn?

Training results and frozen validation answer different questions.



Run	Wall minutes	Macro decisions	Games	Best training	Final dev mean
Scale 1	25.0	9,450	72	304	162.6
Scale .01	25.0	8,508	63	300	162.6
Choice only	25.0	2,586	70	296	162.6
Choice + rate .001	25.0	3,556	91	356	128.0

All four arms began with the same informed weights and a deterministic development mean of 162.6 trips. The best trained development candidate averaged 162.6. Operational selection chose output\training\goal-hybrid-20261028\initial.zip. The frozen initial policy was eligible for selection and wins exact ties. Actor and critic weights were verified to change; this alone is not evidence of better play. Reward scaling addressed numerical loss balance; choice-only timing removed policy calls with no alternative action; the last arm increased learning rate tenfold. These are adaptive engineering pilots, not replicated causal estimates.

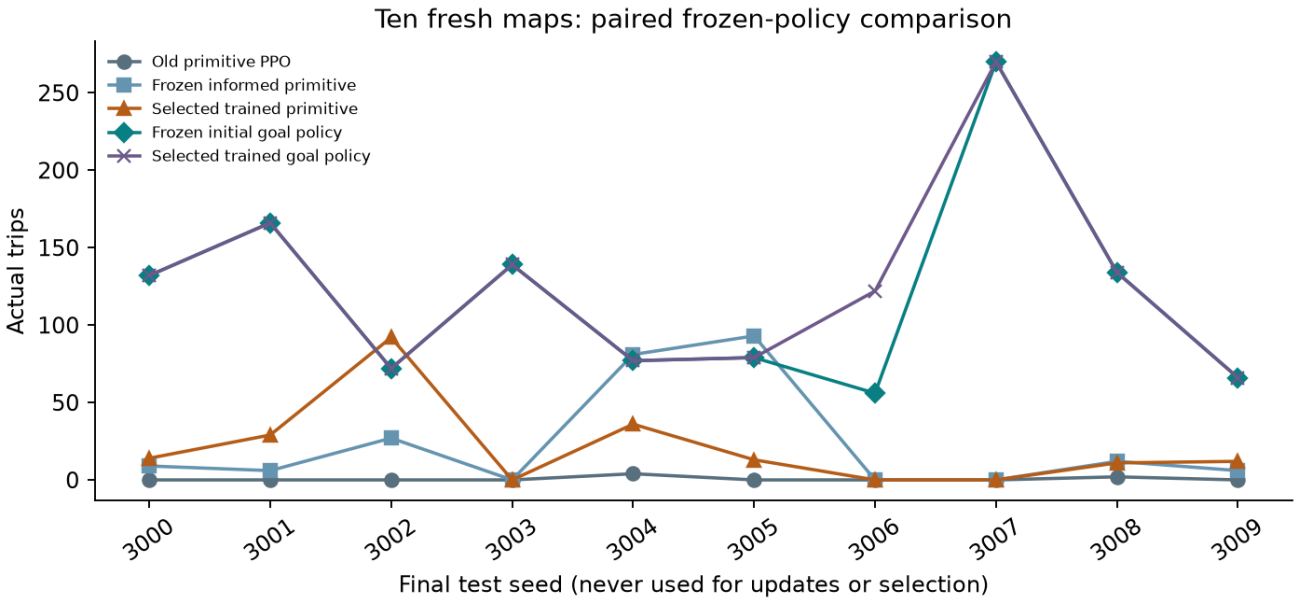
Why games per second changed

There is no deliberate 0.8-games/second limit. Each game lasts a different number of simulated seconds and requires a different amount of routing, serialization and policy work. Better games can take longer to finish. Workers collect experience for one shared policy; they are not separate independently learning models. Path planning adds CPU work, while macro decisions reduce the number of neural choices per game.

These arms start from the same initial checkpoint; they are not consecutive training stages. Scaled rewards multiply numerical learning rewards by 0.01, never actual scores. Choice-only timing folds forced waits into the preceding decision and uses a separately versioned macro reward. Its decision counts are not comparable to five-second-wait counts. Periodic validation consumes learner time; initial/final evaluation is additional. Shared CPU load and changing seed mixes limit comparisons of wall speed and training means.

6 / Fresh-map final test

Candidate selection was recorded before examining seeds 3000-3009. No subsequent optimization uses these results.



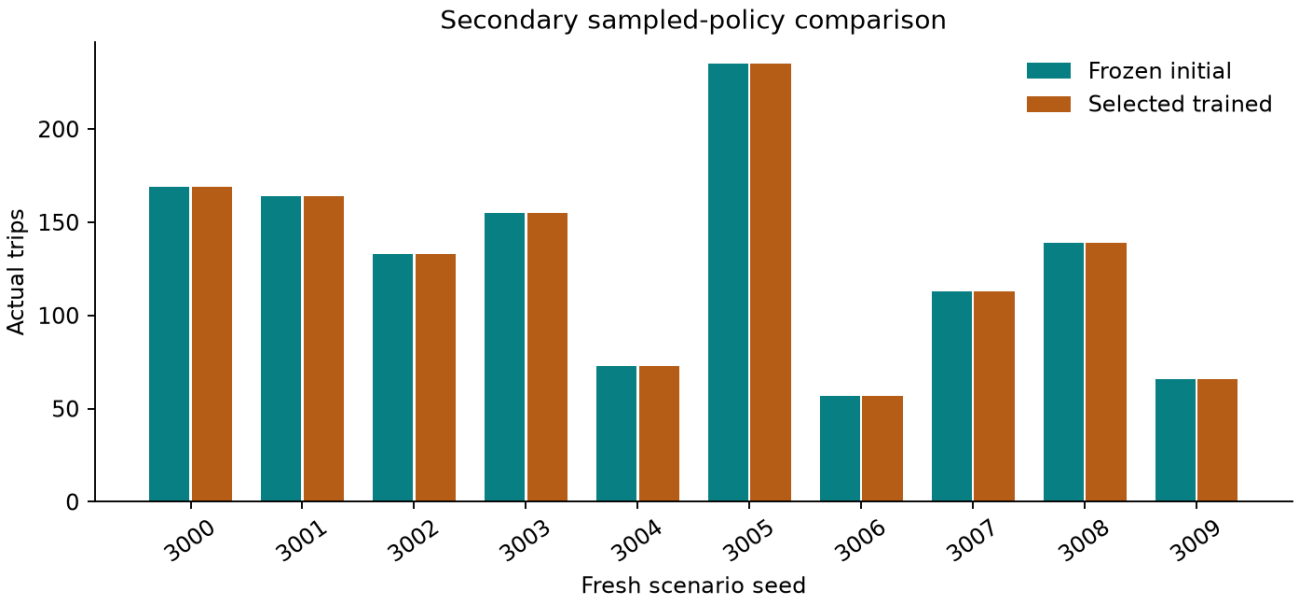
Frozen policy	Mean	Median	Best
Old primitive PPO	0.6	0.0	4
Frozen informed primitive	23.4	7.5	93
Selected trained primitive	20.7	12.5	92
Frozen initial goal policy	119.1	105.5	270
Selected trained goal policy	125.7	127.0	270

Compared with its timing-matched frozen initial policy, the trained goal candidate changed the paired mean by +6.6 trips: Across the ten maps, 1 improved, 0 worsened and 9 tied. Checkpoints were chosen using development maps before these results were opened. No subsequent training or selection used this test set. The old-versus-goal comparison includes engineered route planning and a different action space; it is not a measure of learning alone.

Ten paired scenario seeds, deterministic inference, 3,600 simulated-second external cap, normal failure rules. Frozen model parameters were checked before/after inference and remained unchanged. Training and test city are both Los Angeles; seed separation does not establish transfer to other cities. One training replicate and ten test maps are not evidence of robust generalization or guaranteed future improvement.

7 / Sampled-policy check

Small probability changes can matter even when the most likely action stays the same.



Sampled frozen policy	Mean	Median	Best
Initial goal	130.4	136.0	235
Trained goal	130.4	136.0	235

The frozen initial goal policy averaged 130.4 trips and the selected trained policy 130.4, a change of +0.0. This secondary test samples one trajectory per scenario with Torch seed 43 plus scenario offset, reset after checkpoint loading and before each game. Scenario seeds are paired, but action histories can consume random numbers differently. One sampled trajectory per map is noisy and does not estimate expected performance precisely. It was predeclared before final testing and was not used to select a checkpoint or restart training.

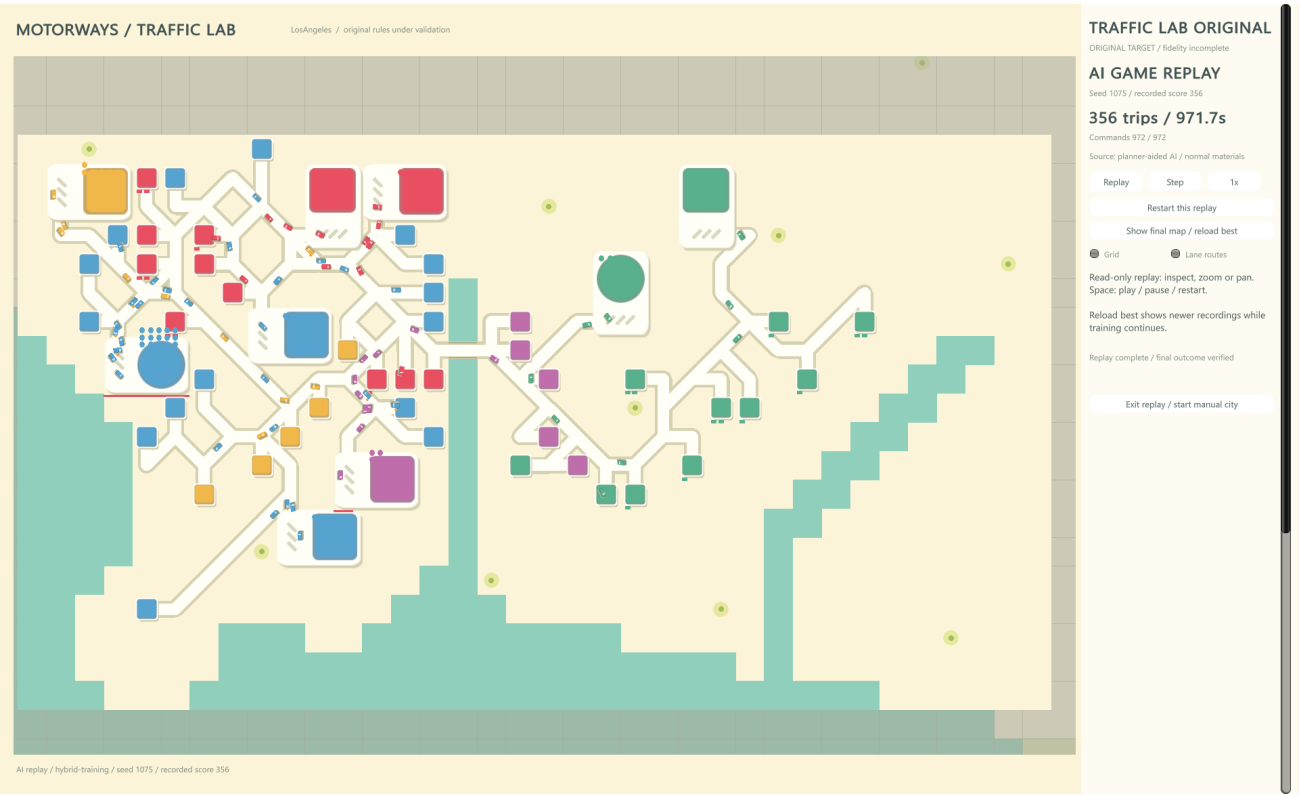
Greedy and sampled play answer different questions

Deterministic evaluation always chooses the currently most likely legal action. Sampled evaluation draws an action from the policy probabilities, as training does. Neither a single sampled win nor an unchanged greedy score proves the whole learning process is successful or broken. Both should be compared against the same starting policy and separate maps.

Each policy gets one sampled game on each of ten scenario seeds, with a 3,600-second cap. The requested RNG seed is applied after loading the checkpoint; reproducibility was checked by repeating a sampled development game and comparing exact ordinary-command recordings. These are inference-only tests: weights remain unchanged.

8 / The best recording

A replay is an exact action history, not proof that the final checkpoint will reproduce that score.



Recorded game	Value
Source	hybrid-training
Seed / actual trips	1075 / 356
Simulated duration / ordinary commands	971.65s / 972
Recording end	Normal failure
Verification	Exact headless/native replay; see final-best-native.log and final-best-headless.txt

Best-game ranking uses actual trips, then survival time, then fewer rejected actions. An episode can span PPO updates and uses sampled actions during training. The exact command recording preserves that game; the checkpoint saved at its end is only a policy snapshot.

Planner-aided and heuristic replays are labeled separately in the native viewer. No material grants, debug construction or future schedule are used. Native/headless agreement validates replay determinism for the tested recordings, not perfect fidelity to the commercial game.

9 / What to do next

Use the measured checkpoint and preserve the comparison, rather than blindly extending an unchanged run.

Preserve both the frozen initial baseline and the development-selected checkpoint. Use the separate goal trainer for further planner-aided experiments and keep the original all-tools trainer as a baseline. Do not assume that more hours on the same seed, more parameters or a higher training best will solve transfer. A next investigation should remove proven unchanged strokes from the router, test capacity-aware route alternatives and meaningful decision credit, and repeat whole training runs with different optimization/sampling seeds and new validation/test partitions. No further run is automatically scheduled.

Run and view

Open PowerShell in C:\Users\moham\mini_motorways. The documented commands in training/QUICKSTART.md use the separate goal trainer for planner-aided experiments and the existing start_training.ps1 for the primitive policy. Every new run needs a fresh output folder. Ctrl+C saves and stops; a stop-file also supports graceful completion.

To inspect the best recording: **.venv/Scripts/python.exe training/watch_best.py --run output/training/goal-rate-20261028**. Add --play to begin paused at the start instead of loading the final map.

Durable evidence and boundaries

The archive contains run configs, source snapshots, summaries, progress/episode logs, frozen evaluations, exact best recordings, selection protocol and SHA-256 manifest. Model weights and generated game binaries remain local. GitHub contains the code, small evidence and PDF on mohammad; main remains the playground.

Relevant checks covered simulator invariants, structured observation boundaries, input migration, actual gradient changes, checkpoint reloads, parallel workers, reward cancellation, no-op masks, shutdown and cross-runtime ordinary-command replays. The rebuilt native player passed the simulator check suite. Full test outcomes and any command corrections are recorded in the experiment log.

Open limits

The experiments are small and adaptive. Development maps influenced engineering decisions, so they are not final tests. The planner uses approximate costs, can encounter ordinary rejection rules and redraws reused path segments. In five frozen initial development games, 438 of 668 accepted draw commands changed no network state; avoiding those is a future executor optimization, not a skill the current policy can learn inside a macro. The goal agent has fewer tools than the original policy. City transfer, replicated learning gains and the 1,000-trip target remain separate questions.

Rebuild: docs/training/build_investigation_report.py using bundled Python with Matplotlib, ReportLab, Pillow and pypdf. The builder reads only archived files and never trains or runs a game. Historical PDFs are preserved. Primary shaping reference: Ng, Harada & Russell (1999), Policy Invariance Under Reward Transformations, ai.stanford.edu/~ang/papers/shaping-icml99.pdf.